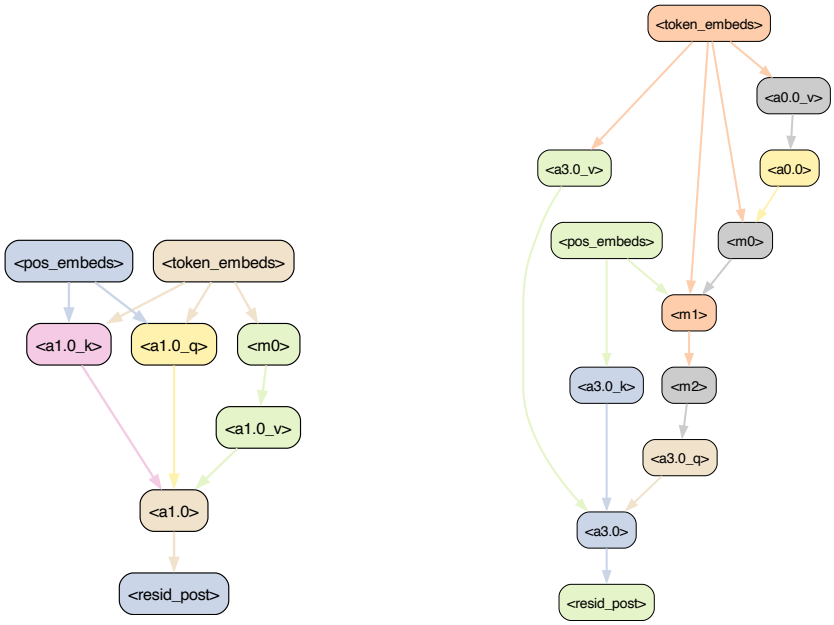

(a) tracr-xproportion canonical circuit (10 edges)

(b) tracr-reverse canonical circuit (15 edges)

Figure 21: The canonical circuits for finding the proportion of 'x' in the input and reversing lists. ACDC recovers these perfectly using zero activations (Table 3 and Figs. 9 and 10).

distributions - in this case they are new lists. We calculate the L2 distance between the one-hot vectors for the recovered list and the true reversed list. For the corrupted dataset, we again let $(x'_i)_{i=1}^n$ be a random permuation of $(x_i)_{i=1}^n$ with no fixed points. Again, at the neuron level a perfect graph is recovered, with the minimal components required to reverse lists (Figure 21b).

J Induction task: details and qualitative evidence

In Section 4.2 we use 40 sequences of 300 tokens from a filtered validation set of OpenWebText (Gokaslan et al., 2019). We filter the validation examples so that they all contain examples of induction — subsequences of the form "$A, B, \ldots, A, B$", where A and B are distinct tokens. We only measure KL divergence for the model's predictions of the second B tokens in all examples of the subsequences $A, B, \ldots, A, B$.

We use both zero activations and corrupted activations to compare ACDC and the other methods. To use ACDC with zero activations, we apply one change to the procedure described in Section 3: instead of setting activations of edges not present in the subgraph to the activations on a corrupted dataset, we set their value equal to 0. We describe how we adapt the methods from Section 4.1 to be used with both zero activations and corrupted activations in Appendix D.1 for SP and Appendix D.2 for HISP.

Our induction experiments were performed on a 2-layer, 8-head-per-layer attention only transformer trained on OpenWebText (Gokaslan et al., 2019). The model is available in the TransformerLens (Nanda, 2022) library.[12] We follow Appendix C of Goldowsky-Dill et al. (2023) for the construction of the dataset of induction examples.

The computational graph has a total of 305 edges, and in Figure 4 we only show subgraphs with at most 120 edges.

[12]The model can be loaded with `transformer_lens.HookedTransformer.from_pretrained(model_name = "redwood_attn_2l", center_writing_weights = False, center_unembed = False)` (at least for the repository version of the source code as of 23rd May 2023)

When iterating over the parents of a given node (Line 4 in Algorithm 1), we found that iterating in increasing order of the head index was important to achieve better results in Figure 4. Similar to all experiments in the work, we iterate in decreasing order of layers, so overall we iterate over head 1.0, 1.1, ... then 1.7, then 0.0, 0.1, ...

An example of a circuit found in the process is given in Figure 6.

K Gendered pronoun completion: qualitative evidence

Mathwin et al. (2023) aim to isolate the subgraph of GPT-2 small responsible for correctly gendered pronouns in GPT-2 small. They do this by studying prompts such as "So Dave is a really great friend, isn't" which are predicted to finish with " he". For that they used ACDC. This presents an example of a novel research project based on ACDC. The result of applying the ACDC algorithm (threshold $\tau = 0.05$) is shown in Figure 22.

The computational subgraphs generated by ACDC on the gendered pronoun completion task show that MLP computations are more important than attention head computations in this task than in the IOI task (Appendix F.2). Early, middle and late layer MLPs have important roles in the subgraph. For example, MLPs 3 and 5 are the important components at the name position (which must be used to identify the correct gender) as they have multiple incident edges: the MLP 7 at the "_is" position has the most incoming connections of any node in the graph, and the late layer MLPs 10 and 11 have the largest direct effect on the output. MLP 7's importance at the "_is" position is an example of a discovery that could not have been made with simpler interpretability tools such as saliency maps. This was early evidence of the summarization motif (Tigges et al., 2023).

ACDC's output shows that the important internal information flow for predicting the correct gender has three steps. Firstly, Layer 0 attention head 0.4 and MLP0 use the name embedding which they pass (through intermediary MLPs) via key- and value-composition (Elhage et al., 2021) to attention heads 4.3 and 6.0. Secondly, heads 4.3 and 6.0 attend to the name position to compose with 0.4 and MLP0. Finally, through value-composition with attention heads 6.0 and 4.3 (via MLP7), the outputs of 10.9 and 9.7 output the expected gendered completion to the output node. Mathwin et al. (2023) then verified that indeed in a normal forward pass of GPT-2 Small, 0.4 has an attention pattern to itself at the name token, attention heads 4.3 and 6.0 attend to the previous name token, and 10.9 and 9.7 attend to the ' is' token. They also perform path patching experiments on intermediate nodes to provide further evidence of the importance of the pathway through the ' is' token.

We used the dataset of $N = 100$ examples from Mathwin et al. (2023) . The corrupted dataset was a set of prompts with a similar structure to the sentence "That person is a really great friend, isn't", following the authors' approach.

We defined a computational graph that featured nodes at the specificity of attention heads split by query, key and value vectors, and further split by token position (where the tokens are present in the nodes in Figure 22). From the input sentence 'So Sarah is a really nice person, isn't', we chose to add nodes representing the model internal operations at the tokens "_Sarah", "_is", "_person", "_isn" and "'t", while other positions were grouped together as in Mathwin et al. (2023) . The resulting subgraph can be found in Figure 22.

L Reset Network Experiments

Our **reset network** experiment setup is motivated by the concern that interpretability explanations may not accurately represent the reasoning process behind models' predictions (Jacovi and Goldberg, 2020). This is particularly relevant to work on subnetworks as empirically some subnetworks in models with randomized weights do not accurately represent such reasoning (Ramanujan et al., 2020).

To this end, we study the task-specific metrics on models with permuted weights (which we call **reset networks** (Zhang and Bowman, 2018; Cao, Sanh, and Rush, 2021)) and verify that the circuit recovery algorithms perform worse on these models that do not have underlying algorithms. Specifically, we create the reset network by permuting the head dimension of each layer's Q, K, V matrices, and each MLP's bias term. This disrupts the functionality of the subject model, without changing many facts about the distribution of the activations (e.g. the average magnitude at every layer). In our experiment

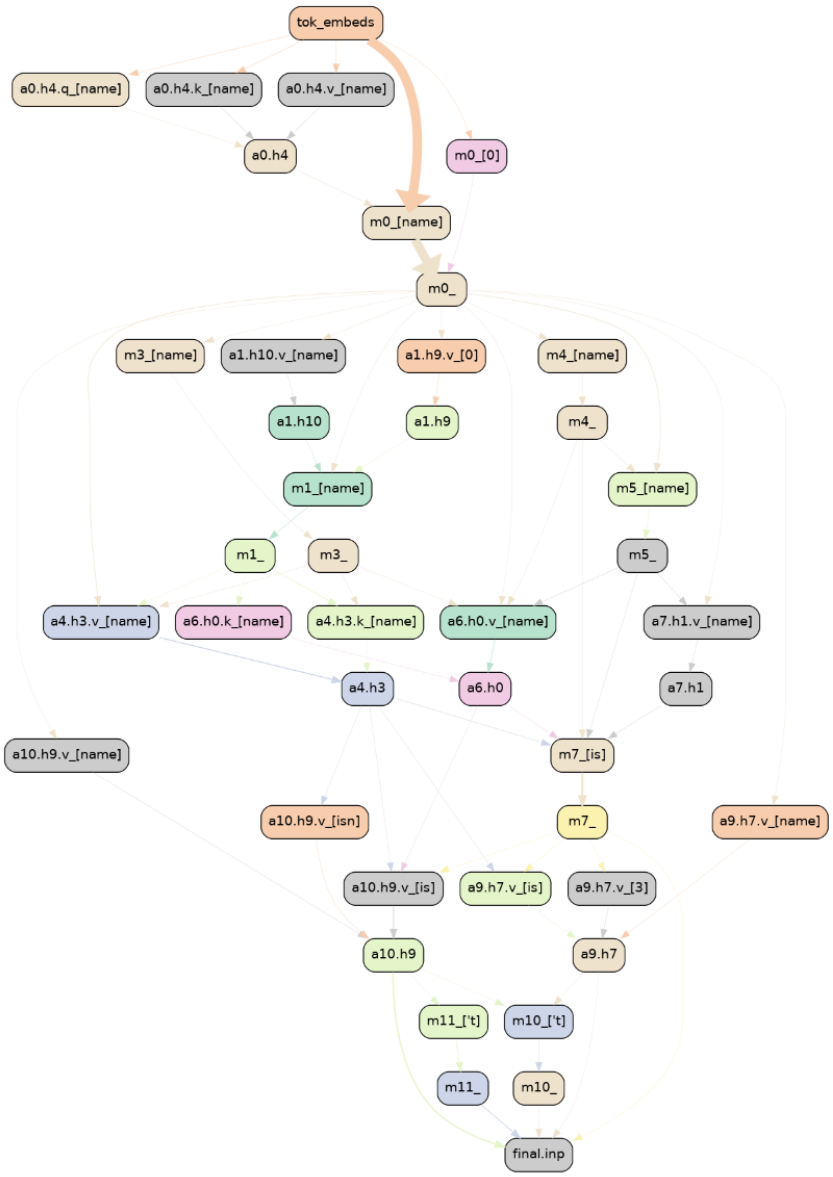

Figure 22: Gendered pronoun completion circuit found by ACDC.